

PROFESSIONAL EXPERIENCE

Kaen Kogashi

Personal Details

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Career Summary

- NEC Soft, Ltd. (currently NEC Solution Innovators, Ltd.) — Apr. 2009 to Mar. 2017
- OMRON Corporation — Jul. 2017 to Nov. 2022
- Mitsubishi Electric Corporation — Dec. 2022 to Present

Core Strengths

Research Areas: Multiple human and object 3D reconstruction, human-object interaction detection, world-action models, tactile dexterous manipulation, and robotics dataset development.
Tools: PyTorch, PyTorch 3D, OpenCV, Qwen, Cosmos3, RGB-D camera systems, CUDA, Shadowhand.

Education

Period	Degree	University
2019-Present	Ph.D. in Intelligence Science and Technology	Kyoto University
2018-2019	Deep Learning Course	University of Tokyo
2003-2009	Bachelor’s and master’s Degrees, Computer Science	Gunma University

Professional Experience

Mitsubishi Electric Corporation (Dec. 2022 – Present)

Period	Project / Responsibilities	Phase / Scope	Environment	Team / Role
Sep. 2025 – Present	Transferred overseas to Mitsubishi Electric Research Laboratories (MERL) in the Boston, U.S. Research on 3D reconstruction of multiple objects and humans, as well as manipulation of deformable objects using a dexterous hand robot with tactile sensors. Application example: sushi making.	Full cycle: concept planning, proof-of-principle model establishment, and publication.	PyTorch; Linux (GPU); Deep Learning; Hardware: Shadow Hand	Sole algorithm researcher; collaboration with more than 10 people.
Dec. 2022 – Aug. 2025	Research on 3D reconstruction of multiple objects and humans.	Full cycle: concept planning, proof-of-principle model establishment, and publication.	PyTorch; Linux (GPU); Deep Learning, Hardware: Multiple Kinect camera system	Sole algorithm researcher; collaboration with more than 10 people.

OMRON Corporation (Jul. 2017 – Nov. 2022)

Period	Project / Responsibilities	Phase / Scope	Environment	Team / Role
Apr. 2021 – Nov. 2022	Prototype development for work activity recognition technology. Objective: realize an industry-first AI system for real-time operator guidance in factory automation. The system was designed to visualize daily variation in operator task times and improve productivity and quality through proficiency support. Mid- to long-term applications included anomaly-work detection and real-time next-task guidance.	End-to-end prototype development, from project planning through proof-of-principle model establishment.	PyTorch; Linux (GPU); Deep Learning	Project leader. Managed collaboration across three sites: a partner startup in Silicon Valley, OMRON Keihanna Innovation Center, and an internal development division.

Apr. 2019 – Mar. 2021	Assigned to OMRON Keihanna Innovation Center; research on human action recognition algorithms.	Action-recognition algorithm research.	PyTorch; Linux (GPU); Deep Learning	Sole algorithm researcher; collaboration with more than 10 people.
Jul. 2017 – Mar. 2019	AI sensor prototype development. Developed a prototype that embedded a machine-learning engine into a general-purpose sensor.	End-to-end prototype development, from project planning through proof-of-principle model establishment.	Python; Windows (CPU); Machine Learning: Binary Tree	Project leader (3 members; more than 10 people). Managed collaboration across a partner startup, Keihanna Innovation Center, and the internal sensor development division.

NEC Solution Innovators, Ltd. (Apr. 2009 – Mar. 2017)

Period	Project / Responsibilities	Phase / Scope	Environment	Team / Role
Jul. 2016 – Jan. 2017	Development of an internal BMS (Business Management System).	Requirements definition; basic/detailed design; programming.	Windows; VBA	2 members / Member
Feb. 2016 – Jun. 2016	Defense wireless communication system. Details omitted due to confidentiality obligations.	Requirements definition; basic design.	Windows; C++	20 members / Member
Jul. 2015 – Jan. 2016	iTEC customization for electricity-market deregulation. Modified the existing iTEC system to support electricity trading.	Basic/detailed design; programming; unit testing.	Windows; Apache; Tomcat; Java; Oracle	20 members / Member
Mar. 2015 – Jun. 2016	Investigation and resolution of existing bugs in pachinko-machine systems for Universal Entertainment.	Bug investigation and resolution.	Linux; Java; C++	10 members / Member
Dec. 2014 – Feb. 2015	Bug investigation and resolution for a Honda car-navigation system for the North American market. Also served as a bridge engineer with a development site in China.	Bug investigation and resolution; bridge engineering.	Linux; Java; Embedded Android	30 members / Member
Jul. 2014 – Nov. 2014	Circle K Sunkus integration with Rakuten Points. Responsible for point handling during product returns.	Basic/detailed design; programming; unit/integration testing.	Windows; C#; VBA	20 members / Member
Feb. 2014 – Jun. 2014	DENSO HEMS (Home Energy Management System).	Configuration management; static analysis; code verification; repository construction.	Windows; Apache; Tomcat; Java; PostgreSQL	30 members / Leader; schedule, quality, and configuration management
Jan. 2014 – Feb. 2014	NESRefine: support for the HEMS China factory version.	Basic/detailed design; programming; unit/integration testing.	Windows; Apache; Tomcat; Java; PostgreSQL; FIAP	3 members / Member
Oct. 2013 – Dec. 2013	Global BEMS: Chinese-language support for a Building Energy Management System.	Basic/detailed design; programming; unit/integration testing.	Windows; Apache; Tomcat; Java; PostgreSQL; FIAP	4 members / Member
Sep. 2013 – Oct. 2013	GUTP (Green University of Tokyo Project): prototype development.	Environment setup; basic/detailed design; programming; unit/integration testing.	Windows; Apache; Tomcat; PHP; PostgreSQL; FIAP	3 members / Member
Aug. 2013 – Sep. 2013	FamilyMart development: development of newly released functionality.	Environment setup; basic/detailed design; programming; unit/integration testing.	Windows; C#	5 members / Member
Jun. 2013 – Jul. 2013	Next-generation UI R&D: prototype development for an air-traffic-control simulation system.	Environment setup; basic/detailed design; programming.	Windows; C#	3 members / Member
May 2013 – Jun. 2013	Tokyo Disneyland ticketing-system update: pre-development investigation of the existing ticketing system.	Investigation of existing system only.	—	Team size unknown / Member
Mar. 2013 – May 2013	BEMS setup: server-side setup for BEMS.	Environment setup.	Windows; Java	3 members / Member
Oct. 2012 – Feb. 2013	HEMS integration-test support.	Integration testing.	—	20 members / Member
Jul. 2012 – Oct. 2012	Modification of Ome City resident-record management system in response to legal amendments. Customized an existing package and coordinated specifications with city-	Environment setup; basic/detailed design; programming; unit testing.	Windows Server 2008; Java	3 members / Member

	hall staff.			
Apr. 2010 – Jul. 2012	Maintenance and development of NEPCELL, NEC's newspaper composition system, used by organizations including Yomiuri Shimbun, Niigata Nippo, and Sankei Shimbun. Responsible for maintenance and development of new functionality.	Bug investigation/resolution; environment setup; basic/detailed design; programming; unit/integration testing.	Linux; C	2 members / Member
Apr. 2009 – Mar. 2010	New employee training. Received training in a wide range of platforms and programming languages and was indirectly involved in development activities.	Training.	Various	Trainee

Qualifications, Awards, and Patents

Internal Award	OMRON Corporation Global Gold Award (Gold Prize), FY2018
External Award	University of Tokyo Deep Learning Course — Excellent Student Award, FY2019
Patents	More than 10 patents related to sensor technologies.

Publications

Kaen Kogashi, Haodi Hu, Toshiaki Koike-Akino. “TacSushi: Tactile-Grounded World-Action Modeling for Dexterous Sushi Manipulation.” IEEE International Conference on Robotics and Automation (ICRA), 2027 (submitted).

Kaen Kogashi, Anoop Cherian, “Structured Affordance-Guided Modeling of Complex 3D Multi-Human-Object Interactions”, MVA (Machine Vision and Applications) Journal, 2027 (submitted).

Kaen Kogashi, Anoop Cherian, Meng-Yu Jennifer Kuo. “MMHOI: Modeling complex 3D multi-human multi-object interactions.” IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026 oral.

Kaen Kogashi, Shohei Nobuhara, Ko Nishino. “SAN: Structure-aware attention network for dyadic human relation recognition in images.” Multimedia Tools and Applications Journal, 2024.

Kaen Kogashi, Shohei Nobuhara, Ko Nishino. “Dyadic human relation recognition.” IEEE International Conference on Multimedia and Expo (ICME), 2022 oral.

Kaen Kogashi, Yang Wu, Shohei Nobuhara, Ko Nishino. “Human–object interaction detection with missing objects.” Image and Vision Computing Journal, 2021.

Kaen Kogashi, Yang Wu, Shohei Nobuhara, Ko Nishino. “Human–object interaction detection with missing objects.” Machine Vision and Applications Conference, 2021 oral.